ECE 3337: Signals & Systems Analysis, Class Worksheet 21

The purpose of this worksheet is to develop familiarity with the Fourier transform by deriving a set of transforms that are listed in Table 5-6 of your textbook. Please derive the following forward and inverse Fourier transforms. The correct answers are in Table 5-6, so your goal should be to practice your ability to perform these derivations correctly.

1. (50 points) Calculate the Fourier transforms for the following signals:

$x(t) = \delta(2t + T)$
$\mathbf{x}(\mathbf{t}) = rect(\frac{3t+2}{5\tau})$
$x(t) = t^{-1} e^{-at} u(t)$
$x(t) = \sin(\omega_0 t)\, u(t)$
$x(t) = e^{-at} \sin(\omega_0 t)\, u(t)$

2. (50 points) Calculate the inverse Fourier transforms for the following signals:

$\hat{X}_1(\omega) = 2$
$\hat{X}_2(\omega) = e^{-j1.5\omega}\left[\frac{\pi}{2}\delta(\omega) + \frac{1}{j\omega}\right]$
$\hat{X}_3(\omega) = \frac{1}{a + j\omega}$
$\hat{X}_4(\omega) = \pi\delta(\omega) - \frac{\pi}{2}[\delta(\omega - 2\omega_0) + \delta(\omega + 2\omega_0)]$
$\hat{X}_5(\omega) = \frac{\omega_0}{(a - j\omega)^2 + \omega_0^2}$